

INTERNSHIP SUMMARY REPORT

on the activities carried out during the internship

Host organisation S.A. "Orange Systems" _____

Department (unit) _____

Faculty of *Computers, Informatics, and Microelectronics* _____

Study programme *Software Engineering* _____

Mode of study *full time* _____

Year of study: 3 _____ **Group:** FAF-243 _____

Internship period: from 01.09.2026 _____ **to** 26.09.2026 _____

Activities carried out and results obtained:

Activities carried out:

During the four-week internship in September 2026, I worked on the backend, the cryptography and the visual side of SecretShare. On the backend, I built the FastAPI endpoints and their routing, including validation, rate limiting and error handling. I set up Redis for ephemeral secret storage and PostgreSQL for device keys and the audit log. I also implemented the browser-side hybrid encryption with the Web Cryptography API (AES-256-GCM for the secret and RSA-OAEP for wrapping the key), and I styled the website.

Results obtained:

I delivered a working API in which secrets are stored only as encrypted envelopes, with an expiry in Redis. I implemented the PostgreSQL data model, including the append-only audit log that stores only an identifier prefix. The encryption runs entirely in the browser, so the server never sees plaintext. I also gave the interface a consistent custom style without third-party UI libraries. The backend passed the automated test suite, with 176 API and security tests in the latest pipeline run.

Difficulties encountered during the internship:

The hardest challenge for me was setting up the API, because it was new to me and I found the file structure difficult to manage. Most of the difficulties I ran into came from the fact that I was not using any design patterns at the start, so responsibilities were mixed together and the code became harder to change as it grew. I only learned about design patterns later, thanks to our mentor, and they helped me organise the backend into clearer layers. This also made it easier to fix problems such as the wrong recipient destroying a secret, identifiers appearing in the access log, and the missing audit table in the preview databases.

Conclusions and recommendations:

The internship taught us to build security in from the start and to test it end to end in production-like environments. Its main drawback was the short duration, which forced us to postpone features such as multi-factor authentication, so we suggest a slightly longer programme or an earlier start on infrastructure setup. The company was good and provided a supportive environment for reaching the internship's goals. Our mentor was very helpful, was genuinely intrigued by our project, and gave us good tips and feedback that improved our work.

Date of completion: 26 / 09 / 2026

Signature of the student: Poiată Călin